

Zhaochen James Xu

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Research interests

Robotic Locomotion, Physics of Living System, Robophysics

Education

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|-------------|--|
| 2024 – now | Georgia Institute of Technology
PhD in Quantitative Bioscience, School of Physics
MS in Robotics/Computer Science (in progress)
GPA:3.9/4.0
Advisor: Prof. Daniel Goldman |
| 2022 – 2024 | Columbia University
MS in Biomedical Engineering
Concentration: Neural Engineering, <i>GPA: 3.8/4.0</i> |
| 2018 – 2022 | University College London
BEng in Electrical Engineering
Minor in Biomedical Engineering, <i>Upper Second-Class Honour</i>
Santander Global Scholarship |
| Summer 2019 | Stanford University
Summer exchange
Intensive program major: Computer Science |

Research experience

May 2023 – now [Crab Lab](#) , Georgia Institute of Technology
Advisor: Daniel I. Goldman

- **Tactile Sensing for Multi-Legged Robots:** Led the experimental development and validation of a robust, compliant antenna that provides tactile feedback during locomotion over rugged terrain. Integrated sensing with robot hardware and feedback control, resulting in a first-author paper accepted to IEEE/RSJ IROS 2026.
- **Multi-Terrain and Surface-Swimming Robots:** Designed and tested undulatory robots with morphable, directionally compliant legs for terrestrial and water-surface locomotion. Characterized gait, body compliance, drag, and water–air interface dynamics using robophysical experiments and resistive force theory, contributing to work published in *Physical Review Applied* and presented at APS Global Physics Summit and SICB.
- **Robust Locomotion in Complex Terrain:** Developed hardware and control architectures for multi-legged elongate robots and investigated geometric, probabilistic, and contact-based gait control on noisy and rugged landscapes. This research contributed to publications in *IEEE Transactions on Robotics*, ICRA, and ongoing work on agile asymmetric locomotion.

Dec 2022 – Jan 2024 [LIINC](#), Columbia University
Advisor: Paul Sajda

- Executed non-invasive electrophysiology studies, focusing on signal processing across multi-modal brain imaging on human.
- Coded an fMRI preprocessing pipeline utilizing Nipype on Python, achieving capabilities in skull-stripping, smoothing, coregistration, and slice-timing correction.
- Deployed a signal processing pipeline via MNE on Python for EEG data, investigating the coupling between pupil and neuronal population dynamics with an emphasis on pre-stimulus alpha wave responses.

Sep 2021 – May [Lombardo's Lab](#), **University College London**

2022 *Advisor: Antonio Lombardo*

- Contributed to the design, fabrication, and test of miniaturized micro-/milli-meter biosensors, specifically targeting their application in early disease detection via breath biopsy.
- Applied techniques including COMSOL multi-physics simulation, vector network analyzer, and scanning microwave microscopy to assess biosensor usability and resolution due to different dielectric property.
- The final microfluidic channel developed has the capability of differentiate two types of volatile organic compounds (VOCs) with ± 10 dB of S parameter magnitude.

Teaching experience

Spring 2023

BMEN4002: Quantitative Physiology II

Professor. Barclay Morrison

- Graded assignments, quizzes, and exams, provided constructive feedback and maintained accurate records of grades.
- Conducted twice-a-week office hour sessions to reinforce concepts and guide students through problem sets.
- Hosted review sessions before midterms and final for 61 students to help prepare them for exams.

Industry experience

Mar 2021 - Aug
2021

TikTok China

Data Engineer

Interned at TikTok Ads' local small business team (SMB), working on advertising data analysis; collaborated across 20 cities, delivered reports, and modeled quotas using Python, Hive SQL, VBA.

Mentorship and leadership

Aug 2023 – now

BME department Mentor, Columbia BME

Guided and supported incoming Master's students in the BME program, offering insights into curriculum planning, research opportunities, and campus resources. Conducted orientation sessions, provided one-on-one advice.

Sep 2018 – Sep
2019

Department Representative, UCL Engineering

Acted as a liaison between students and faculty; Assisted an average of 5 students per week in resolving academic and administrative issues; Worked with a team of 8 faculty and staff members to plan and executed 3 department events.

Technical skills

Robotic Sim

MuJoCo, Isaac Sim/Isaac Lab

Data and ML

MATLAB, Python (NumPy, Pandas, PyTorch, Tensorflow, MNE), MySQL

Hardware Development

Oscilloscope, Analog & digital circuit, Electrophysiology study, NI Multisim, Altium PCB design, Cadence, CST Studio, COMSOL, Arduino, Antenna, Wireless, RF Engineering, Verilog, 3D Printing, Signal processing, Motion tracking, AutoCAD

Software Development

Git, GitHub, Bash Command, Linux/Unix, Docker

Publications

- Paper**
- [1] **Z. J. Xu**, J. He, D. Aydan, M. Taylor, T. Wang, J. Lin, W. Dyer, D. I. Goldman. “A Robust Antenna Provides Tactile Feedback in a Multi-Legged Robot,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, accepted, 2026. arXiv preprint arXiv:2603.07795.
 - [2] B. J. Walker, C. Moreau, T. L. Robinson, **Z. J. Xu**, D. I. Goldman, E. A. Gaffney. “Tunable Asymmetric Swimming in Biflagellate Microswimmers,” *Journal of The Royal Society Interface*, accepted for publication, 2026. arXiv preprint arXiv:2602.17521.
 - [3] J. Habala, G. B. Margolis, T. Wang, P. Bhatt, J. He, N. Naeem, **Z. Xu**, P. Agrawal. “Agile Asymmetric Multi-Legged Locomotion: Contact Planning via Geometric Mechanics and Spin Model Duality,” *arXiv preprint arXiv:2602.09123*, 2026.
 - [4] J. Lin, T. Wang, B. Chong, M. Fernandez, **Z. Xu**, D. I. Goldman. “Optimal Swimming with Body Compliance in an Overdamped Medium,” *Physical Review Applied*, **25**, 054068, 2026. doi:10.1103/pbkh-py5h.
 - [5] B. Chong, J. He, D. Irvine, T. Wang, E. Flores, D. Soto, J. Lin, **Z. Xu**, D. I. Goldman. “Robust Control for Multi-Legged Elongate Robots in Noisy Environments,” *arXiv preprint arXiv:2506.15788*, 2025.

[6] J. He, B. Chong, J. Lin, **Z. Xu**, H. Bagheri, E. Flores, D. I. Goldman. “Probabilistic Approach to Feedback Control Enhances Multi-Legged Locomotion on Rugged Landscapes,” *IEEE Transactions on Robotics*, 2025.

[7] M. Iaschi, B. Chong, T. Wang, J. Lin, J. He, D. Soto, **Z. Xu**, D. I. Goldman. “Addition of a Peristaltic Wave Improves Multi-Legged Locomotion Performance on Complex Terrains,” *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2925–2931, 2025.

[8] J. He, B. Chong, **Z. Xu**, S. Ha, D. I. Goldman. “Learning to Enhance Multi-Legged Robot on Rugged Landscapes,” *arXiv preprint arXiv:2409.09473*, 2024.

Abstract

[9] **Z. J. Xu**, J. Lin, T. Wang, D. I. Goldman. “Morphable Legs with Directional Compliance for Multi-Terrain Multilegged Robot Swimming,” *APS Global Physics Summit*, 2026.

[10] **Z. Xu**, L. Treers, B. Chong, J. Lin, K. Diaz, D. Goldman. “Resistive Force Theory Modelling of Locomotion at the Water–Air Interface,” *Integrative and Comparative Biology*, vol. 66, p. 317, 2026.

[11] **Z. J. Xu**, B. Chong, J. Lin, A. Lauria, T. Wang, D. I. Goldman. “Rapid Legged Locomotion in an Isotropic Frictional Environment,” *SMT 2025*, 2025.

[12] **Z. J. Xu**, S. Scarpinato, B. Chong, L. Treers, K. Diaz, D. I. Goldman. “Water Surface Swimming Dynamics in a Lightweight Undulatory Robot,” *SMT 2025*, 2025.

[13] J. He, B. Chong, **Z. Xu**, E. Flores, D. Soto, D. Goldman. “Tactile Feedback Enhances Multi-Legged Locomotion on Rugged Landscapes,” *APS March Meeting Abstracts*, G38.005, 2024.

[14] J. He, B. Zhong, E. Flores, **Z. Xu**, D. Soto, D. Goldman. “Tactile Feedback Enhances Multi-Legged Locomotion on Rugose Terrain,” *Integrative and Comparative Biology*, vol. 64, pp. S220–S221, 2024.

[15] B. Chong, D. Luo, T. Wang, G. Margolis, **Z. Xu**, M. Iaschi, P. Agrawal, M. Soljačić, D. I. Goldman. “Geometry of Contact: Contact Planning for Multi-Legged Robots via Spin Models,” *APS March Meeting Abstracts*, G38.002, 2024.